MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 07 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. The area of the triangle formed by the lines $x = 3$, $y = 4$ and $x = y$ is: [1]

(A) 1/2 sq. unit (B) 1 sq. unit (C) 2 sq. units (D) None of these

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 1/2 sq. unit** [0.5 Mark]

Justification: Pairwise intersections of $x = 3$, $y = 4$, and $x = y$ give vertices $(3, 4)$, $(3, 3)$, and $(4, 4)$. [0.5 Mark]

This is a right-angled triangle at $(3, 4)$ with horizontal leg $= 4 - 3 = 1$ and vertical leg $= 4 - 3 = 1$. Hence Area $= \frac{1}{2} \times 1 \times 1 = \mathbf{\frac{1}{2}}$ sq. unit.

Q2. Which of the following is represented by the system $3x + 4y = 5$ and $6x + 8y = 7$? [1]

(A) Parallel lines (B) Intersecting lines (C) Coincident lines (D) Perpendicular lines

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Parallel lines** [0.5 Mark]

Justification: $\frac{a_1}{a_2} = \frac{3}{6} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{4}{8} = \frac{1}{2}$, and $\frac{c_1}{c_2} = \frac{-5}{-7} = \frac{5}{7} \neq \frac{1}{2}$. Hence **parallel lines.** [0.5 Mark]

Q3. A linear equation in two variables $ax + by + c = 0$ (a, b not both zero) graphically represents: [1]

(A) a circle (B) a straight line (C) a parabola (D) a hyperbola

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) a straight line** [0.5 Mark]

Justification: Every solution of a first-degree equation $ax + by + c = 0$ lies on a unique **straight line.** [0.5 Mark]

Q4. For what value of k does the system $8x + 5y = 9$ and $kx + 10y = 15$ have no solution? [1]

(A) 8 (B) 16 (C) 4 (D) 20

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 16** [0.5 Mark]

Justification: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies \frac{8}{k} = \frac{5}{10} = \frac{1}{2} \implies k = \mathbf{16}$ (and $\frac{9}{15} = \frac{3}{5} \neq \frac{1}{2}$). [0.5 Mark]

SECTION B: Assertion-Reason Questions MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. Assertion (A): The linear equations $x - 2y = 0$ and $3x + 4y - 20 = 0$ intersect at $(4, 2)$.
Reason (R): A point lies on both lines if and only if its coordinates simultaneously satisfy both equations. [1] *Directions: (A) Both true & R explains A; (B) Both true, R doesn't explain A; (C) A true, R false; (D) A false, R true.*

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Testing $(4, 2)$: $4 - 2(2) = 0$ (satisfies Eq. 1) and $3(4) + 4(2) - 20 = 12 + 8 - 20 = 0$ (satisfies Eq. 2). [0.5 Mark]

Since the slopes differ ($1/3 \neq -2/4$), $(4, 2)$ is their unique intersection point. Both (A) and (R) are true and (R) explains (A).

Q6. Assertion (A): The line represented by $y - 3 = 0$ is parallel to the x-axis.
Reason (R): Any equation of the form $y = c$ ($c \neq 0$) represents a straight line parallel to the x-axis. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: $y - 3 = 0 \iff y = 3$, which matches $y = c$ ($c = 3 \neq 0$), a horizontal line 3 units above the x-axis. [0.5 Mark]

Both (A) and (R) are true and (R) explains (A).

SECTION C: Very Short Answer Questions MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. Solve for x and y: $\sqrt{2}x + \sqrt{3}y = 0$ and $\sqrt{3}x - \sqrt{8}y = 0$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Substituting x from Equation (i) into (ii) [1 Mark]

From $\sqrt{2}x + \sqrt{3}y = 0$, we obtain $x = -\frac{\sqrt{3}}{\sqrt{2}}y$ — (iii)

Substituting into $\sqrt{3}x - \sqrt{8}y = 0$ (noting $\sqrt{8} = 2\sqrt{2}$):

$$\sqrt{3}\left(-\frac{\sqrt{3}}{\sqrt{2}}y\right) - 2\sqrt{2}y = 0 \implies -\frac{3}{\sqrt{2}}y - 2\sqrt{2}y = 0 \implies -\left(\frac{3+4}{\sqrt{2}}\right)y = 0 \implies -\frac{7}{\sqrt{2}}y = 0$$

Step 2: Solving for y and x [1 Mark]

Hence $y = 0$. Putting $y = 0$ in (iii) gives $x = -\frac{\sqrt{3}}{\sqrt{2}}(0) = 0$.

$$\boxed{\mathbf{x = 0, \quad y = 0}}$$

Q8. Find the value of c for which the system $cx + 3y = 3$ and $12x + cy = 6$ has no solution. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Formula / Theorem Used: No Solution Condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Step 1: Equating Ratios [1 Mark]

$\frac{c}{12} = \frac{3}{c} \neq \frac{-3}{-6} = \frac{1}{2} \implies c^2 = 36 \implies c = \pm 6.$

Step 2: Checking Inequality Condition [1 Mark]

At $c = 6$: $\frac{6}{12} = \frac{1}{2} = \frac{c_1}{c_2}$ (infinitely many solutions, rejected). At $c = -6$: $\frac{-6}{12} = -\frac{1}{2} \neq \frac{1}{2}$ (valid).

$$\boxed{\mathbf{c = -6}}$$

Q9. Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Forming Equations [1 Mark]

Let length $= l$ m and width $= b$ m. Half perimeter $= l + b = 36$ — (i), and $l = b + 4$ — (ii).

Step 2: Solving for b and l [1 Mark]

Substituting (ii) into (i): $(b + 4) + b = 36 \implies 2b = 32 \implies b = 16$ m. Thus $l = 16 + 4 = 20$ m.

$$\boxed{\text{Length} = \mathbf{20}\text{ m}, \qquad \text{Width} = \mathbf{16}\text{ m}}$$

SECTION D: Short Answer Questions — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q10. A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid Rs. 27 for a book kept for seven days, while Susy paid Rs. 21 for the book she kept for five days. Find the fixed charge and the charge for each extra day. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let fixed charge for first 3 days be Rs. x and extra charge per day be Rs. y.

Step 1: Setting Up Linear Equations [1.5 Marks]

Saritha ($7 - 3 = 4$ extra days): $x + 4y = 27$ — (i)

Susy ($5 - 3 = 2$ extra days): $x + 2y = 21$ — (ii)

Step 2: Solving by Elimination [1.5 Marks]

Subtracting (ii) from (i): $(x + 4y) - (x + 2y) = 27 - 21 \implies 2y = 6 \implies y =$ Rs. 3.

Substituting in (ii): $x + 2(3) = 21 \implies x = 21 - 6 =$ Rs. 15.

$$\boxed{\text{Fixed Charge} = \text{Rs. } \mathbf{15}, \qquad \text{Extra Day Charge} = \text{Rs. } \mathbf{3}\text{/day}}$$

Q11. Meena went to a bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. Meena received 25 notes in all. Find how many notes of Rs. 50 and Rs. 100 she received. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let number of Rs. 50 notes be x and Rs. 100 notes be y.

Step 1: Formulating Equations [1.5 Marks]

Total notes: $x + y = 25$ — (i)

Total amount: $50x + 100y = 2000 \implies x + 2y = 40$ — (ii)

Step 2: Solving for y and x [1.5 Marks]

Subtracting (i) from (ii): $y = 40 - 25 = 15$. Putting $y = 15$ in (i): $x = 25 - 15 = 10$.

$$\boxed{\mathbf{10}\text{ notes of Rs. }\mathbf{50} \quad \text{and} \quad \mathbf{15}\text{ notes of Rs. }\mathbf{100}}$$

Q12. Solve for x and y: $\dfrac{x}{a}+\dfrac{y}{b}=2$ and $ax-by=a^2-b^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Clearing Denominators [1 Mark]

Multiplying $\frac{x}{a}+\frac{y}{b}=2$ by ab $(a,b\neq 0)$: $bx+ay=2ab$ — (i), and $ax-by=a^2-b^2$ — (ii).

Step 2: Eliminating y and Finding x, y [2 Marks]

Multiplying (i) by b and (ii) by a: $b^2x+aby=2ab^2$ — (iii) and $a^2x-aby=a^3-ab^2$ — (iv).

Adding (iii) and (iv): $(a^2+b^2)x=a^3+ab^2=a(a^2+b^2)\implies x=a$ (since $a^2+b^2>0$).

Substituting $x=a$ into $\frac{x}{a}+\frac{y}{b}=2$: $1+\frac{y}{b}=2\implies \frac{y}{b}=1\implies y=b$.

$$\boxed{\mathbf{x=a,\quad y=b}}$$

SECTION E: Long Answer / HOTS Question MATHS MANTHAN | (1 × 5 = 5 Marks)

Q13. Two years ago, a father was five times as old as his son. Two years later, his age will be 8 years more than three times the age of the son. Formulate the pair of linear equations and find their present ages. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let father's present age be x years and son's present age be y years.

Step 1: Translating Age Relations [2 Marks]

2 years ago: $x-2=5(y-2)\implies x-2=5y-10\implies x-5y=-8$ — (i)

2 years later: $x+2=3(y+2)+8\implies x+2=3y+14\implies x-3y=12$ — (ii)

Step 2: Solving Equations by Elimination [2 Marks]

Subtracting (i) from (ii): $(x-3y)-(x-5y)=12-(-8)\implies 2y=20\implies y=10$ years.

Step 3: Finding Father's Present Age [1 Mark]

Substituting $y=10$ into (i): $x-5(10)=-8\implies x=50-8=42$ years.

$$\boxed{\text{Father's Age} = \mathbf{42}\text{ years},\qquad \text{Son's Age} = \mathbf{10}\text{ years}}$$

Q14. CASE STUDY: Public Municipal Library Book Lending System

A city central library promotes reading by operating an affordable lending window. The borrowing rules specify a fixed fee of Rs. x for the first 3 days and an additional fee of Rs. y for each day thereafter. Radhika borrowed a science encyclopedia for 7 days and paid Rs. 27. Aakash borrowed a history reference manual for 5 days and paid Rs. 21. [4]

(i) Formulate the pair of linear equations in x and y representing Radhika's and Aakash's borrowings. [1]

(ii) Determine the fixed charge for the first three days (x). [1]

(iii) Determine the additional daily charge per extra day (y). [1]

(iv) If Priya borrows a novel for 10 days, calculate the total amount she will have to pay. [1]

— OR —

If Aman paid an amount of Rs. 45, find how many days he kept the borrowed book. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Formula / Theorem Used: Total Fee $= x + (d-3)y$ for $d \geq 3$ days

Sub-part (i): Linear Equations [1 Mark]

Radhika ($7 - 3 = 4$ extra days): $x + 4y = 27$ — (i) Aakash ($5 - 3 = 2$ extra days): $x + 2y = 21$ — (ii)

Sub-part (ii): Fixed Charge for First 3 Days (x) [1 Mark]

Subtracting (ii) from (i): $2y = 6 \implies y = 3$. Putting $y = 3$ in (ii): $x + 6 = 21 \implies \boxed{\mathbf{x = \text{Rs. } 15}}$.

Sub-part (iii): Additional Charge per Extra Day (y) [1 Mark]

From the subtraction step above: $\boxed{\mathbf{y = \text{Rs. } 3/\text{day}}}$.

Sub-part (iv): Priya's Borrowing for 10 Days [1 Mark]

Extra days $= 10 - 3 = 7$. Total fee $= x + 7y = 15 + 7(3) = 15 + 21 = \boxed{\text{Rs. } \mathbf{36}}$.

Sub-part (iv) [OR Option]: Days Kept by Aman for Rs. 45 [1 Mark]

$15 + (d-3)(3) = 45 \implies 3(d-3) = 30 \implies d - 3 = 10 \implies d = \boxed{\mathbf{13} \text{ days}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★